Claude Code + Ollama Setup Guide

Run Claude Code entirely on free, local models through Ollama — no Anthropic account needed, nothing leaves the machine.

One-time setup (once per machine)

macOS / Linux

```
curl -fsSL https://claude.ai/install.sh | bash   # installs Claude Code
ollama pull llama3.2:3b                           # downloads the model, ~2GB
```

(Install Ollama itself first, from ollama.com/download, or `brew install ollama`)

Windows (PowerShell)

```
irm https://claude.ai/install.ps1 | iex
ollama pull llama3.2:3b
```

(Install Ollama itself first, from ollama.com/download)

Check both worked: `ollama --version` and `claude --version`.

Start a session

Every time, one line — works from any folder, any machine, no Anthropic login required:

macOS / Linux

```
ANTHROPIC_AUTH_TOKEN=ollama ANTHROPIC_BASE_URL=http://localhost:11434
ANTHROPIC_API_KEY="" claude --model llama3.2:3b
```

Windows (PowerShell)

```
$env:ANTHROPIC_AUTH_TOKEN="ollama";
$env:ANTHROPIC_BASE_URL="http://localhost:11434"; $env:ANTHROPIC_API_KEY="";
claude --model llama3.2:3b
```

Swap `llama3.2:3b` for any other pulled model — run `ollama list` to see exact tags.

What to expect

These are normal, not errors:

- A one-line warning that the model "isn't described by this version's model catalog" — expected for any non-Anthropic model, harmless.
- The first message after starting (or after a few idle minutes) is slower — the model is loading into memory. Later messages in the same session are much faster.
- A one-time "trust this folder?" prompt the first time `claude` runs in a new folder — accept it.

Choosing a model

`llama3.2:3b` (used above) is the safe default for a classroom — small download, runs on modest laptops, supports tool use.

Model	Download	Good for
`llama3.2:1b`	1.3GB	Slowest machines, quick demos
`llama3.2:3b`	2.0GB	Recommended default
`qwen3.8`	~17GB	Stronger reasoning; needs a capable laptop and patience

Pull any of these the same way — `ollama pull <name>` — then use that exact name after `--model`.

Troubleshooting

Symptom	Fix
`Please run /login` · `401 Invalid bearer token`	The three `ANTHROPIC_...` variables weren't set for that command — retype the full line from **Start a session**, in the same terminal window
Nothing happens for a long time	Normal on the first message — wait 1–2 minutes for the model to load, especially larger ones
`ollama: command not found`	Ollama isn't installed, or the terminal needs restarting after install
Model name not recognized	Run `ollama list`, use the exact tag shown (including the `:tag` suffix)